

Cerebrotendinous Xanthomatosis

EPIDEMIOLOGY

The prevalence of cerebrotendinous xanthomatosis (CTX) is approximately 1 per 50,000.

ETIOLOGY AND PATHOGENESIS

The genetic abnormality results in incomplete oxidation of cholesterol into bile acids in the liver due to mutations in mitochondrial C27-steroid 26-hydroxylase (CYP27A1). A by-product of this defect is increased synthesis of the sterol cholestanol.

CLINICAL FINDINGS

In CTX, xanthomata develop in the tendons and brain. CTX causes severe neurologic manifestations. Patients with juvenile cataracts should undergo testing for CTX, which can be diagnosed by elevated plasma cholestanol.

PROGNOSIS AND CLINICAL COURSE

Patients with CTX are at increased risk for coronary heart disease (CHD).

TREATMENT

Treatment consists of replacement of the bile acid pool with oral chenodeoxycholic acid. This therapy reduces cholestanol synthesis and has been reported to alleviate neurologic symptoms. HMG-CoA reductase inhibitors and LDL apheresis are also reported as possible treatments. A combination of LDL apheresis, chenodeoxycholic acid, and a HMG-CoA reductase inhibitor more effectively reduces serum cholestanol than LDL apheresis alone.

Tangier Disease

ETIOLOGY AND PATHOGENESIS

Tangier disease is a rare, autosomal-recessive condition caused by defects in the gene *ABCA1*. This protein is responsible for transporting cellular cholesterol to the cell surface where it is delivered to circulating HDL. A defect in the transmembrane transporter causes an accumulation of cholesterol in peripheral tissues due to the inability to perform "reverse cholesterol transport" via HDL particles.

CLINICAL FINDINGS

Cutaneous and Systemic Lesions

Tangier homozygotes accumulate cholesterol esters in tonsils, spleen, lymph nodes, bone marrow, thymus, and intestinal mucosa. These accumulations most commonly produce yellow/orange hyperplastic tonsils, mild lymphadenopathy, and splenomegaly.

Laboratory Findings

Patients with Tangier disease have very low plasma HDL cholesterol and apolipoprotein A1 (apo-A1) levels.

PROGNOSIS AND CLINICAL COURSE

Older patients with Tangier disease often develop coronary heart disease (CHD).

High-Density Lipoprotein Metabolic Defects: Apolipoprotein A1 Deficiency

ETIOLOGY AND PATHOGENESIS

Apo-A1 is the major apolipoprotein in HDL. Mutations in apo-A1 can reduce HDL synthesis and lead to HDL deficiency.

CLINICAL FINDINGS

Patients with defects limited to apo-A1 synthesis may develop plane xanthomata.

PROGNOSIS AND CLINICAL COURSE

It is uncertain whether these patients are prone to premature CHD; however, they do not typically develop fulminant atherosclerosis. In some cases, there is a concomitant defect in synthesis of apo-C3 and apo-A4. When these additional defects are present, complex abnormalities in lipoprotein metabolism occur, and affected patients are at increased risk for premature CHD.

Dysproteinemias

ETIOLOGY AND PATHOGENESIS

Diffuse plane xanthomata have been reported in association with multiple myeloma, monoclonal gammopathy, cryoglobulinemia, leukemia, lymphoma, eosinophilic granulomatosis, and rheumatoid arthritis. When diffuse xanthomas occur in monoclonal gammopathies, the M protein is usually immunoglobulin G

(IgG). Xanthomas are formed due to complexes between the paraprotein and lipoproteins. Cellular uptake of these complexes by macrophages leads to the development of plane xanthomata.

CLINICAL FINDINGS: CUTANEOUS LESIONS

A number of dysglobulinemias or paraproteinemias are accompanied by diffuse plane xanthomata that typically appear on the neck, trunk, and flexural areas.